Ex:No:6 Error correction at Data link layer.

Aim:

Write a program to implement error detection and correction using hamming code concept. Make a test run to input data stream and verify error correction.

Error correction at Data link layer:

* Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is transmitted from the sender to the receiver.

Create a sender program:

⇒ Input to sender file should be a text of any length. Program should convert the text to binary.

⇒ Apply hamming code concept on the binary data and add redundant bits to it.

⇒ Save this output in a file called channel.

Create a receiver program:

⇒ Receiver program should read the input from channel file.

⇒ Apply hamming code on a binary data to check for errors.

⇒ If there is an error, display the position of the error.

⇒ Else remove the redundant bits and convert the binary data to ascii and display the Output.

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Code : (sender).

```
#include <stdio.h>
#include <string.h>
void hanning_code (char data, char code[13])
{
  int d[9];
  int p1, p2, p4, p8;
  for (int i=0; i<8; i++){
    d[i+1] = (data >> (7-i)) & 1;
  }
  code[3] = d[1];
  code[5] = d[2];
  code[6] = d[3];
  code[7] = d[4];
  code[9] = d[5];
  code[10] = d[6];
  code[11] = d[7];
  code[12] = d[8];
  p1 = code[3] ^ code[5] ^ code[7] ^ code[9] ^ code[11];
  p2 = code[3] ^ code[6] ^ code[7] ^ code[10] ^ code[11];
  p4 = code[5] ^ code[6] ^ code[7] ^ code[12];
  p8 = code[9] ^ code[10] ^ code[11] ^ code[12];
}
int main(){
  file *fp;
  char text[1000];
  char code[13];
  printf("Enter the text: ");
  fgets(text, sizeof(text), stdin);
  fp = fopen("channel.txt", "w");
  if (fp == NULL)
  {
    printf("Error opening channel");
    return 1;
  }
```

```
for(int i=0; i<strlen(text); i++) {
    if(text[i] == '\n')
      break;
    hamming_encode(text[i], code);
    for(int j=1; j<=12; j++) {
      fprintf(fp, "%d", code[j]);
    fprintf(fp, " ");
  }
  fclose(fp);
  printf("Data encoded successfully");
  return 0;
}
```

Code : (receiver).

```
#include <stdio.h>
int main(){
    file *fp;
    char block[20];
    int code[13];
    int data;
    char ch;
    int error_position;
    fp=fopen("channel.txt","r");
    if(fp==NULL){
    printf("Error");
    return 1;
    }
    printf("\n Received data:\n");
    while(fscanf(fp,"%s",block)==1){
      for(int i=1; i<=12; i++){
          code[i]=block[i-1]-'0';
      }
      error_position = p1*1 + p2*2 + p4*4 + p8*8;
```

```
if (error_position == 0){
  printf ("No errors detected");
}
else {
  code [error_position] = code [error_position] ^ 1;
  printf ("Error corrected");
}
data = 0;
data = (data << 1) | code [3];
data = (data << 1) | code [5];
data = (data << 1) | code [6];
data = (data << 1) | code [7];
data = (data << 1) | code [9];
data = (data << 1) | code [10];
data = (data << 1) | code [11];
data = (data << 1) | code [12];
  ch = (char) data;
  printf (" Received character: %c \n", ch);
}
  fclose (fp);
  printf ("Data received successfully. \n");
  return 0;
}
```

Output:

Enter the text: Hello

Data encoded successfully.

Hamming Coded data is stored in channel.txt.

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